AJCR Horizon Status: Why Phase 7 Is Stuck

Human mathematician handoff

24 August 2026

Snapshot: 2026-08-24 10:06 +0800. **Source ledger:** `a58a86aca1`.

This handoff compares Horizon with the supplied execution plan and supervision note. It is written in mathematical stages; Lean file pointers are included only where they help locate the claim.

`MainProjects/Algebraic-Jacobian-Challenge/informal/Lean_Algebraic_Jacobian_Complete_Execution_Plan.pdf`

`MainProjects/Algebraic-Jacobian-Challenge/informal/AJCR_Runs_121_122_123_Supervision_Note_2026-08-07.pdf`

Status at a glance

GREEN — The rank-one chart and the separably-closed Pic0 representer are reported as genuine results.

AMBER — The finite-stage descent construction is present in source, but the critical compiled proof cone is incomplete.

RED — The universal original-field representer and the exact geometric hypothesis needed by the quotient are still absent.

The important correction to the previous handoff is that the family-level rank-one producer is no longer the blocker.

What has landed

The reviewed route is:

```
rank-one Picard/divisor chart
  -> family-level Abel isomorphism
  -> translated rank-one charts over a separably closed field
  -> Pic^0 represented over that field
  -> finite-Galois descent
  -> Pic^0 represented over the original field
  -> Jacobian
```

Stage	Status	Mathematical meaning and pointer
Rank-one chart	**Complete**	The good locus of line bundles and the matching divisor locus are defined for arbitrary test schemes. `Pic0CriticalPath.lean:205`
Family Abel map	**Complete**	The evaluation divisor gives a natural two-sided inverse, rather than only fieldwise uniqueness. `Pic0CriticalPath.lean:618`
Sep-closed cover	**Complete**	Translations of the rank-one chart cover Picard classes after passing to a separably closed field. `Pic0CriticalPath.lean:640`
Sep-closed Pic0	**Complete**	A scheme and a representing universal class exist over the separably closed field; the same pair feeds the current datum objects. `Pic0SepClosedRepresentable.lean:443`

The narrow source audit for this part reports only `propext`, `Classical.choice`, and `Quot.sound`.

Where progress stops

Horizon is now trying to descend the separably-closed representer through a custom finite-stage atlas:

```
finite-stage affine charts and overlap maps
  -> candidate glued scheme P.gluedOver
  -> universal Picard class on that scheme
  -> natural equivalence: maps to P.gluedOver <-> Pic^0 families
  -> projectivity / affine-orbit control for the same scheme
  -> finite-Galois quotient and original-field Pic^0
  -> Jacobian data
```

The candidate scheme is defined in `Pic0FiniteStageGluedOver.lean:1`; the decomposition is in `AJCR.review-plan.p7-galois-descent.yaml`.

Remaining item	Status	What must be shown
Compile the glued scheme	**Build blocked**	A fresh root import must certify the diagram isomorphism, the two overlap projections, and the glued comparison. `GluePackage.olean` appeared on Aug 24; four top artifacts remain absent. `Pic0FiniteStageGluingDiagramIso.lean:276`
Finite-stage representation	**Missing**	Show that maps into `P.gluedOver` are naturally equivalent to families of `Pic0` classes. This is the universal property, not just chart compatibility. `p7-galois-descent.representability.finite-stage.yaml` `DivisorScheme.tex:1846`
Exact-carrier geometry	**Missing**	Prove projectivity or affine-orbit control for this exact glued scheme. Existing quotient results assume this property; they do not produce it. `p7-galois-descent.orbit-affine.exact-carrier.yaml`
Original-field Pic0	**Missing**	Produce the original-field scheme together with its universal representation, rather than a theorem that accepts a representation as an input. `p7-galois-descent.representability.original-field.yaml`
Jacobian handoff	**Blocked downstream**	The final `JacobianData` must use the same original-field scheme and universal class. That handoff does not exist. `Pic0FiniteGaloisJacobianData.lean:1`

The finite-stage compile cone is concentrated in
`Pic0FiniteStageGluePackage.lean:1`
,
`Pic0FiniteStageGluingDiagramIso.lean:1`
,
`Pic0FiniteStageGluingOverlapIsoPreSnd.lean:1`
,
`Pic0FiniteStageGluingOverlapIsoSnd.lean:1`
, and
`Pic0FiniteStageGluedComparison.lean:1`
.

Why Horizon appears stuck

1. **Mathematical and build work are mixed.** After the sep-closed milestone, Horizon split one producer-level descent target into transition models, triple overlaps, ring maps, glue records, base-change maps, and projection lemmas. These are useful prerequisites, but none is the universal property of the candidate scheme.

2. **The critical compile is too expensive for normal iteration.** The GluePackage probe used about 7.1 GB of memory. The first expensive declaration in the next file is `overlapBaseChangeIso_hom_iota`, line 276 of `Pic0FiniteStageGluingDiagramIso.lean`; historical checks reached roughly 610 seconds.

3. **Source labels are being confused with root evidence.** The graph contains `lean_ok` labels for source declarations whose compiled artifacts are absent. The critical root is `Pic0CriticalPath.lean:1`; a meaningful acceptance check needs a fresh native build, a root import, and an axiom check.

4. **Orchestration leaves dead work looking active.** Run 0154 still has metadata marked `running` from `2026-08-23T14:55:00Z`, while Horizon reports a zombie marker and no live Lean process. Its system sessions returned the task to `queued`. Runs 0145 and 0148 repeatedly relaunched the same alignment task; later sessions mostly returned `exit 1` or `queued`. The current reports are:

   `runs/0153/.../report.md`

   `runs/0154/.../report.md`

   `runs/0156/.../report.md`

5. **The candidate carrier is the architectural uncertainty.** The plan says to reuse the AJC finite-Galois quotient/gluing engine. Horizon instead first constructs `P.gluedOver` and then needs a new universal equivalence on it. This may be correct, but it may also be a second descent stack not yet shown to represent the same `Pic0` functor. The generic descent contract is in `blueprint/src/chapters/DivisorScheme.tex:1846`.

The decision point

The next useful human decision is whether `P.gluedOver` is the correct finite-level representer for the reviewed descent theorem. If it is, Horizon needs one bounded producer milestone: a root-imported universal equivalence for that carrier. If it is not, the route should be refactored around the generic AJC descent theorem before more overlap and glue files are added.

Run pattern

Run	Date	Observable result
0145	14 Aug	Eight-round alignment task; later sessions repeatedly returned `exit 1` or `queued`.
0148	14 Aug	Same alignment task relaunched; metadata/audit work, then queued or failed sessions.
0149	14 Aug onward	Audit classified 266 commits as 0 acceptance edges, 82 consumed prerequisites, and 10 conditional consumers.
0153	22 Aug	Compile isolation measured the GluePackage/DiagramIso frontier.
0154	23–24 Aug	Compile repair produced `GluePackage.olean`, but no root-certified downstream top cone; stale `running` metadata remains.
0156	23 Aug	Roadmap decomposition separated build, universal-property, orbit, and original-field gates; no theorem edge was added.

The current task database has 28 done, 3 blocked, 4 queued, 1 running, 1 failed, and 45 cancelled tasks. Open inbox items include 12 conversations, 18 issues, and 11 memories. The dashboard still has an empty `generatedAt`, and the existing status PDFs describe the older Aug 21 state.

BOTTOM LINE — Horizon is not stuck because the rank-one route failed. It is stuck because Phase 7 interleaves a resource-bound custom glue construction with missing universal and geometric producers. The central issue is whether `P.gluedOver` is the right object for the reviewed finite-Galois route.